SECOND REVIEW

# PneumoCare : Pneumonia Classification Using Chest X-Ray Images

**Healthcare Analytics**


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# Project overview

## Problem

Pneumonia is a common respiratory infection that is diagnosed partly from chest X-rays, which need expert reading.

## Aim

Classify paediatric chest X-rays as NORMAL or PNEUMONIA with transfer learning, and report honest, audited metrics.

## Intended users

Healthcare Analytics students and faculty exploring methods. It is an educational prototype, not for clinical use.

### PROJECT OBJECTIVES

1. Train and evaluate a ResNet-18 transfer-learning classifier, with the test set used only once
2. Explain predictions with Grad-CAM and state openly what it does and does not show
3. Implement Tutorials T1–T15 as working, tested modules in one application
4. Keep imaging, EHR, text and survey data separate, and label synthetic data clearly
5. Verify every tutorial workflow through the live API with real datasets

# Progress since Review 1

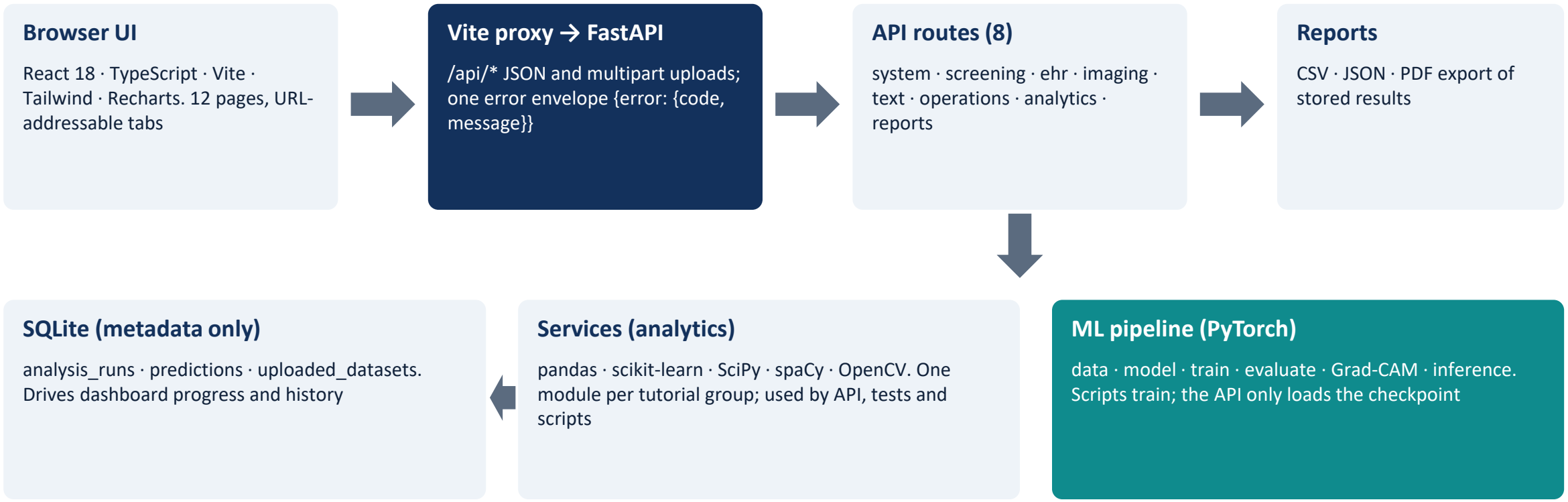
REVIEW 1 · DATA PIPELINE

- Problem definition and objectives
- Data pipeline: loading, splits, 224×224 resize, augmentation, weighted sampling, sanity checks
- Class-imbalance observation
- Tutorials T1–T6 as separate exercises
- No trained model or results yet

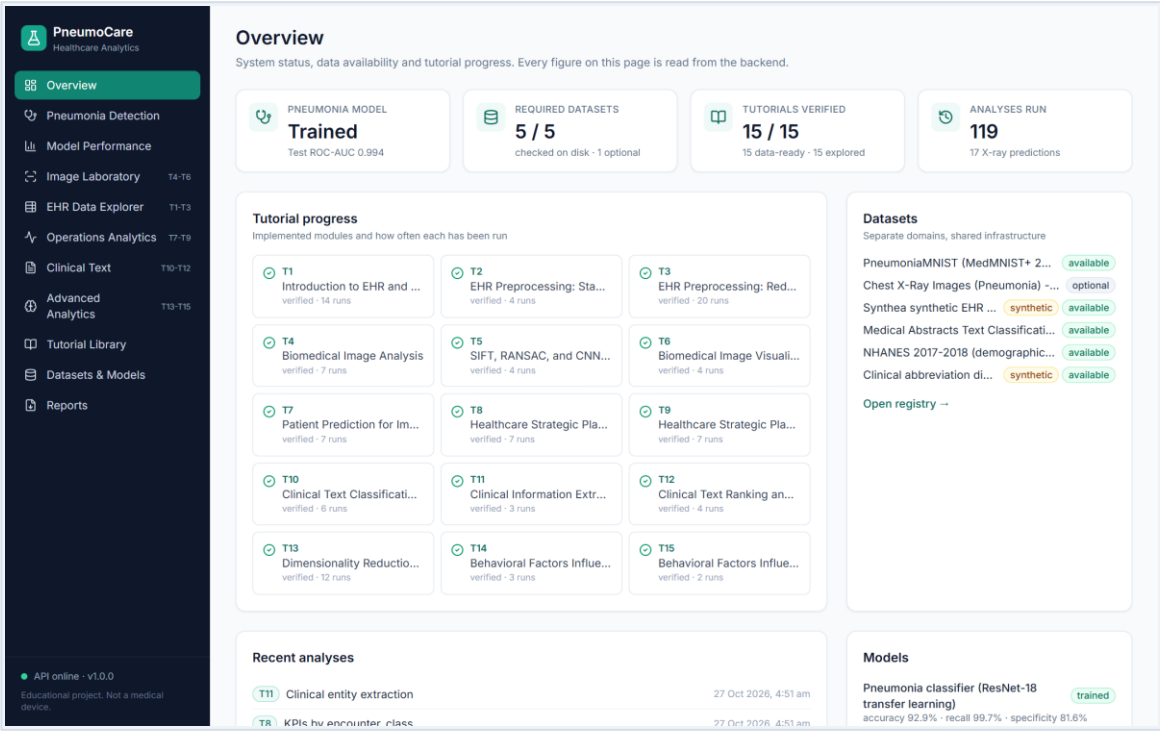
REVIEW 2 · NOW

| | |
|---|---|
| **Model** | ResNet-18 trained (6 epochs, ~39 min CPU); test ROC-AUC 0.994; audit 9/9 |
| **App** | React + FastAPI + SQLite: 12 pages, 8 API routers, analysis history, CSV/JSON/PDF reports |
| **Tutorials** | All 15 implemented in the app; 15/15 verified through the live API (64/64 checks) |
| **Quality** | 85 backend + 15 frontend tests, lint and type check clean |
| **Audit** | Run counts, dataset switching and Grad-CAM claims corrected; segmentation rebuilt; model inventory |

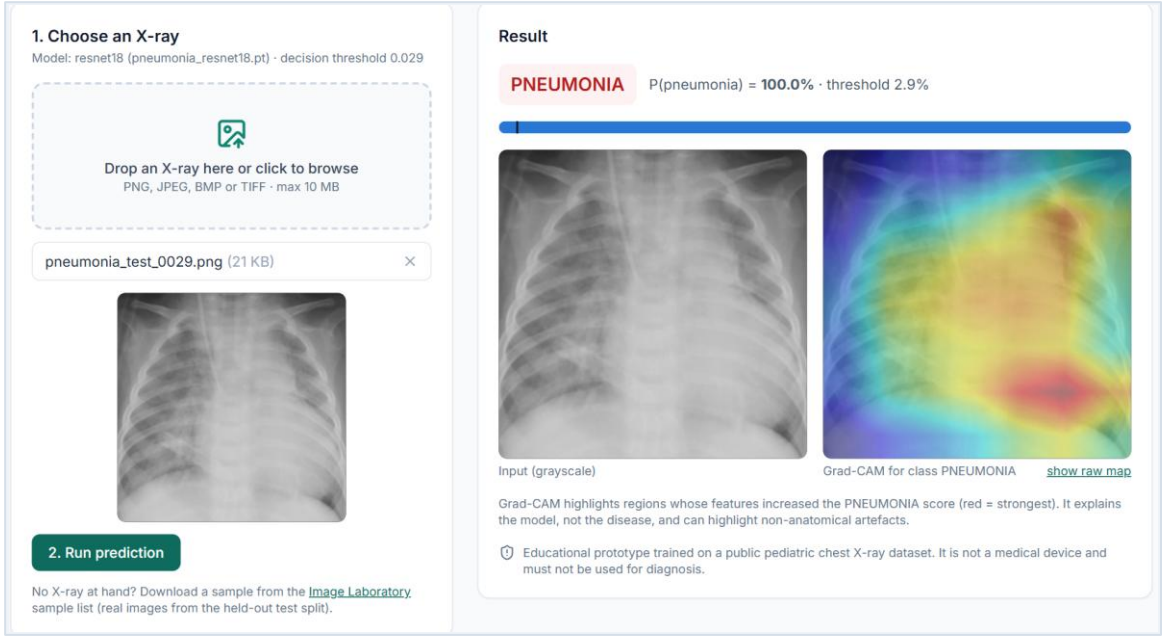
# System architecture



# Application walkthrough

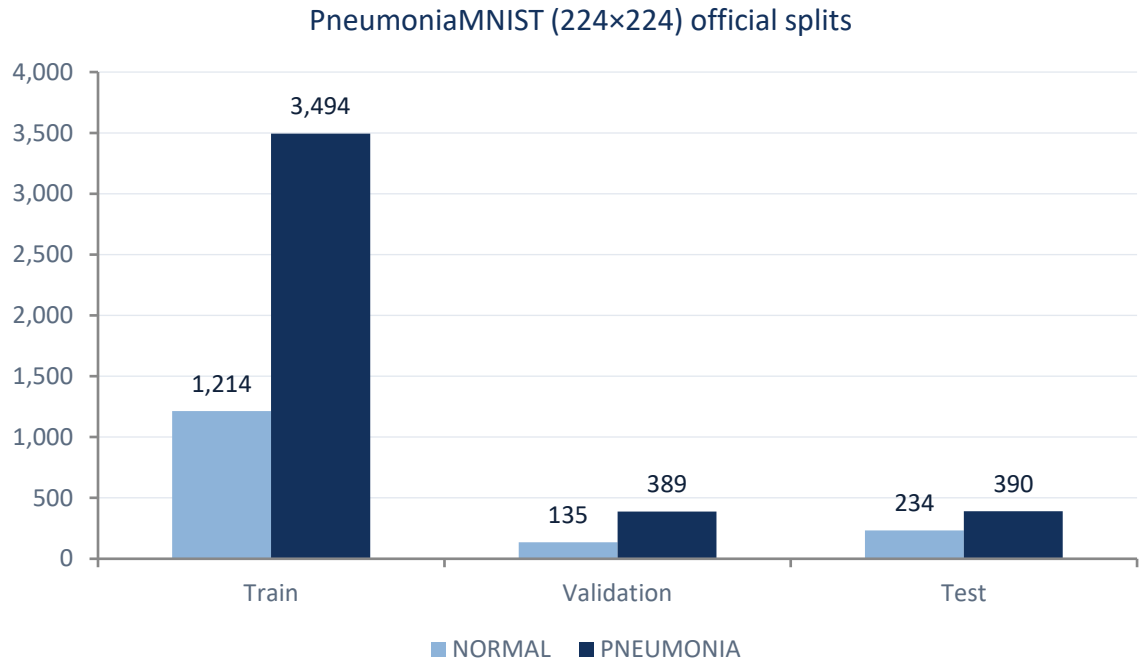



Overview: system status, 5/5 required datasets, 15/15 tutorials verified, recent analyses




Pneumonia Detection: real test X-ray → P(pneumonia), threshold, Grad-CAM overlay. Grad-CAM shows where the model looked, not where disease is

# Dataset and preprocessing




Real, de-identified paediatric X-rays (Kermany et al., 2018) via MedMNIST+, CC BY 4.0.
Pneumonia share: 74.2% train/val, 62.5% test.

**PREPROCESSING (TRAIN | VAL / TEST / INFERENCE)**

| | |
|---|---|
| Grayscale → 3 channels | all splits |
| Resize 224×224 | all splits |
| Random affine ±10°, ±5% shift, 0.9–1.1 scale | train only |
| Brightness / contrast jitter ±15% | train only |
| ImageNet mean/std normalisation | all splits |
| No horizontal flip (heart side must not change) | excluded |

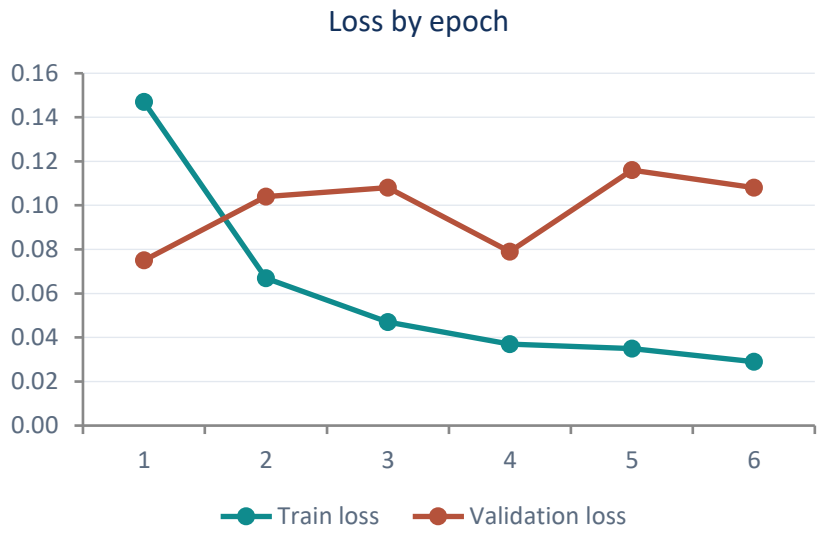
**LEAKAGE CHECK (AVERAGE HASH + PIXEL CONFIRMATION)**

8 exact train–validation duplicates removed from validation (val 524 → 516). 0 duplicates involving the test set, so test metrics are not inflated by leakage. MedMNIST has no patient IDs, so patient-level overlap cannot be checked.

# Model methodology: ResNet-18 transfer learning

Chest X-ray (224×224) → ImageNet ResNet-18 backbone, all layers fine-tuned → Dropout 0.3 → Linear 512→2 → Softmax: P(pneumonia) → Compare with threshold → label → Grad-CAM on layer4

| Setting | Value |
|---|---|
| Loss | Cross-entropy, class weights (NORMAL 1.94, PNEUMONIA 0.67) |
| Optimiser | AdamW, lr 1e-4, weight decay 1e-4, cosine schedule |
| Batch / epochs | 32 / 6 (early stopping, patience 3 on validation ROC-AUC) |
| Model selection | Best validation ROC-AUC (epoch 6: 0.9983) |
| Threshold | Youden J on validation = 0.029 (chosen before testing) |
| Seed / hardware | 42 / CPU, 2,347 s (~39 min) |




Validation ROC-AUC was already 0.996 after epoch 1; training loss keeps falling while validation loss stays flat.

# Model evaluation on the official test set (624 images)

**0.994**
ROC-AUC: ranking quality across all thresholds

**0.930**
Accuracy at the validation threshold 0.029

**0.997**
Sensitivity at 0.029 (1 missed case)

**0.816**
Specificity at 0.029 (43 false alarms)

| Metric | At 0.029 (validation) | At 0.5 (comparison) |
|---|---|---|
| **Accuracy** | 0.930 | 0.968 |
| **Precision (PPV)** | 0.900 | 0.970 |
| **Recall / sensitivity** | 0.997 | 0.979 |
| **Specificity** | 0.816 | 0.949 |
| **F1-score** | 0.947 | 0.974 |
| **ROC-AUC (threshold-free)** | 0.994 | 0.994 |




Rows: actual NORMAL / PNEUMONIA · columns: predicted NORMAL / PNEUMONIA

**Reading the results.** ROC-AUC measures how well the model ranks pneumonia above normal, independent of any threshold; accuracy depends on the chosen threshold. The validation-chosen threshold transfers poorly to the test set, so specificity is low. We report 0.5 for comparison but did not switch to it: choosing a threshold on test data would leak test information.

# How the tutorials fit into PneumoCare

### EHR Data Explorer

**T1 · T2 · T3**

Synthea (synthetic EHR)

### Biomedical Image Lab

**T4 · T5 · T6**

PneumoniaMNIST + trained model

### Pneumonia Detection

**Main project · T6 Grad-CAM**

PneumoniaMNIST

### Operations Analytics

**T7 · T8 · T9**

Synthea encounters (synthetic)

### Clinical Text

**T10 · T11 · T12**

Medical Abstracts (real) + synthetic notes

### Advanced Analytics

**T13 · T14 · T15**

NHANES 2017–2018 (real survey)

**SHARED BY EVERY TUTORIAL**

One FastAPI backend · SQLite analysis history (dashboard progress) · Reports (CSV/JSON/PDF) · Tutorial Library with evidence-based “verified” status · common UI and chart components. Data domains are never merged.

# T7 · Patient Prediction for Improved Healthcare Staffing

*Forecast patient volume and translate it into a staffing (FTE) estimate*

**DATASET** SYNTHETIC DATA

Synthea encounters from 2016: 3,935 encounters, 101 patients, 129 monthly periods

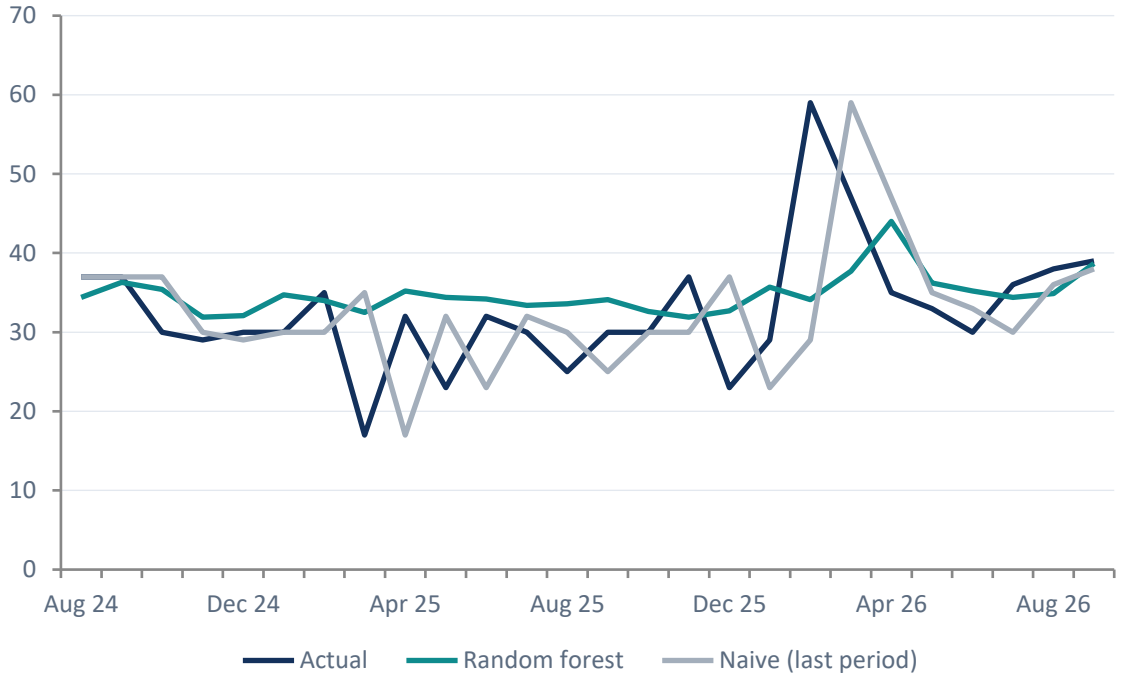
**WORKFLOW**

| INPUT | PROCESSING | OUTPUT |
|---|---|---|
| Monthly encounter counts | Lag + season features, time-ordered split, Ridge & random forest vs baselines | 6-month forecast, workload hours, FTE |

**METHODS IMPLEMENTED**

- Lags 1–3, rolling mean, seasonal sin/cos, trend
- Latest 20% of periods held out (no future leakage)
- FTE = workload h ÷ (160 h × 85% utilisation)

Held-out months: actual vs predicted encounters




**ACTUAL RESULT**

Random forest MAE 5.71 vs naive 6.62 (MAPE 19%). Next month ≈ 39.5 encounters ≈ 0.15 FTE

**KEY LEARNING**

No staffing labels exist, so forecast demand and convert it to workload instead

**IN PNEUMOCARE**

Operations Analytics → Volume & staffing

# T8 · Healthcare Strategic Planning Part 1

*Compute healthcare KPIs, compare groups statistically and follow trends*

**DATASET** SYNTHETIC DATA

Synthea encounters joined with patients, payers and organisations (2016–2026)

| 3,935 | $3,301 | 71.1% | 5.0 d |
|---|---|---|---|
| encounters | avg claim cost | payer coverage | avg length of stay |

## WORKFLOW

**INPUT**
Encounters filtered by start year

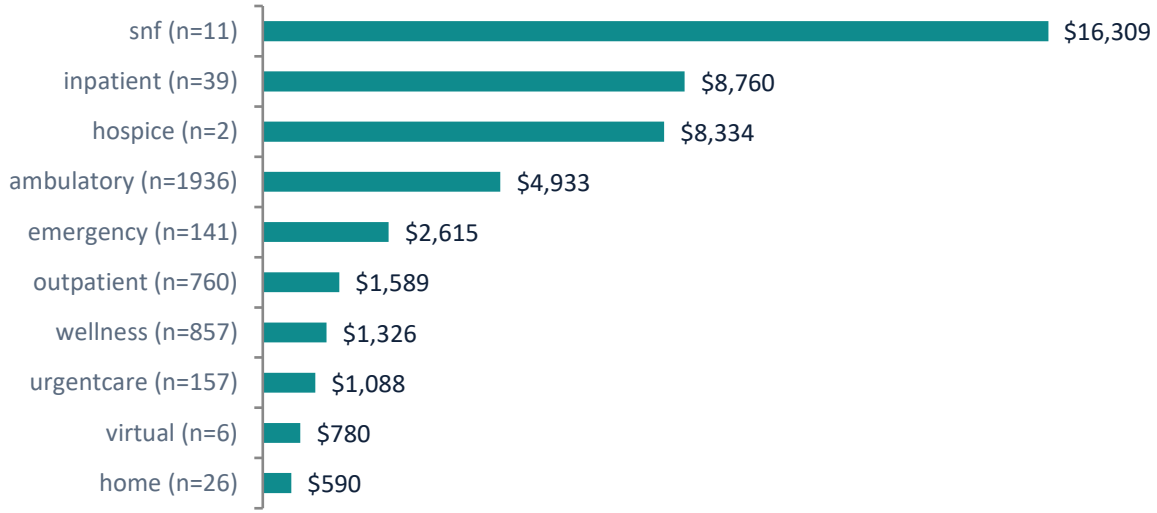
**PROCESSING**
KPIs, group-by, rank-based significance test

**OUTPUT**
KPI cards, group charts, trend lines

## METHODS IMPLEMENTED

- Volume, cost, payer coverage, LOS, 30-day readmission, ED visits / 1,000
- Group by class, year, gender, age, payer or organisation
- Kruskal–Wallis H (Mann–Whitney U for 2 groups): costs are skewed

Average claim cost by encounter class




**ACTUAL RESULT**
Cost differs strongly by class: Kruskal–Wallis H = 307.8, $p = 8.9\times10^{-62}$; SNF and inpatient cost most

**KEY LEARNING**
Skewed cost data need rank-based tests; significance is not practical importance

**IN PNEUMOCARE**
Operations Analytics → KPIs & comparisons

# T9 · Healthcare Strategic Planning Part 2

*Analyse demand patterns and resource utilisation, and compare what-if scenarios*

## DATASET

**SYNTHETIC DATA**

Synthea encounters (2016+); bed capacity is a user assumption (default 5 beds)

## WORKFLOW

**INPUT**

Encounter timestamps and long stays

**PROCESSING**

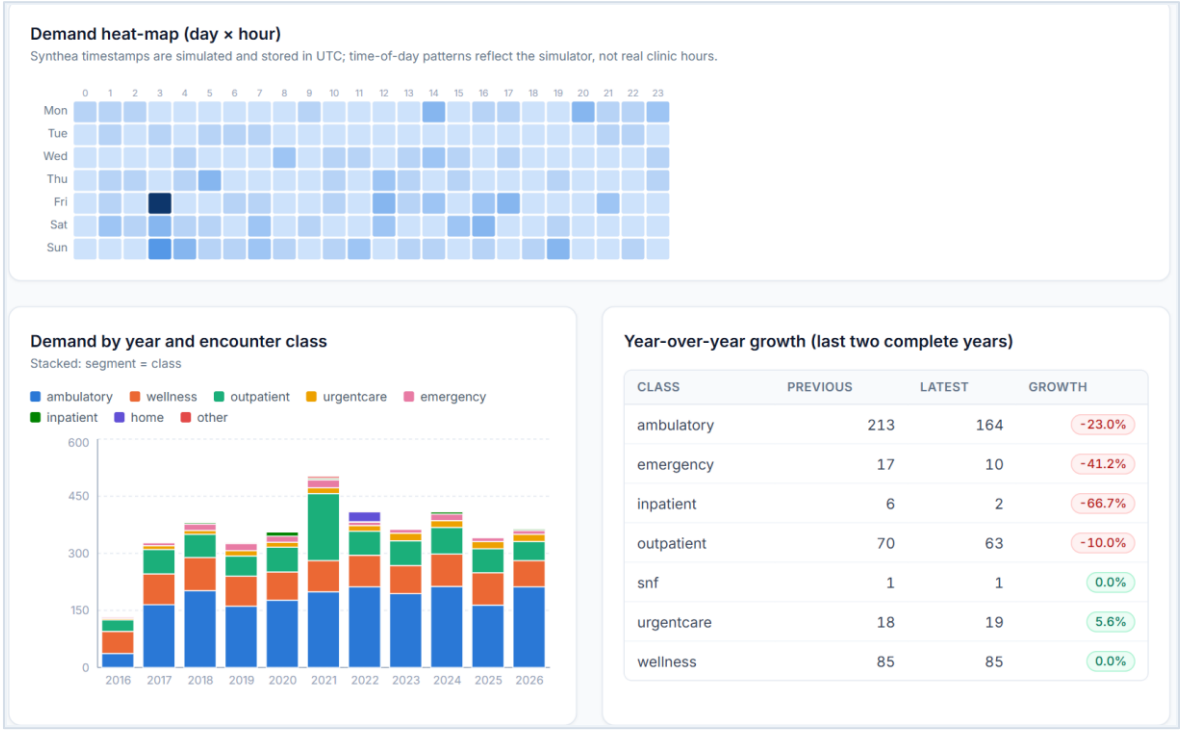
Day×hour crosstab, provider workload, bed occupancy, scenario formulas

**OUTPUT**

Heat-map, utilisation charts, scenario table

## METHODS IMPLEMENTED

- Demand heat-map and demand by year and class
- Bed utilisation = occupied bed-days ÷ (beds × days)
- Scenarios: demand %, staff FTE → utilisation, unmet hours, FTE needed




| CLASS | PREVIOUS | LATEST | GROWTH |
|---|---|---|---|
| ambulatory | 213 | 164 | -23.0% |
| emergency | 17 | 10 | -41.2% |
| inpatient | 6 | 2 | -66.7% |
| outpatient | 70 | 63 | -10.0% |
| snf | 1 | 1 | 0.0% |
| urgentcare | 18 | 19 | 5.6% |
| wellness | 85 | 85 | 0.0% |

## ACTUAL RESULT

Top 10 of 215 providers handle 25% of work. Demand +30% lifts utilisation 63.6% → 82.7%

## KEY LEARNING

Utilisation turns demand into capacity decisions; linear what-ifs ignore queues and variability

## IN PNEUMOCARE

Operations Analytics → Utilisation & scenarios

# T10 · Clinical Text Classification and Clustering

*Preprocess clinical text, classify it with supervised models, and find structure by clustering*

**DATASET** REAL DATA

Medical Abstracts TC Corpus: 11,550 train / 2,888 test abstracts, 5 disease classes

**WORKFLOW**

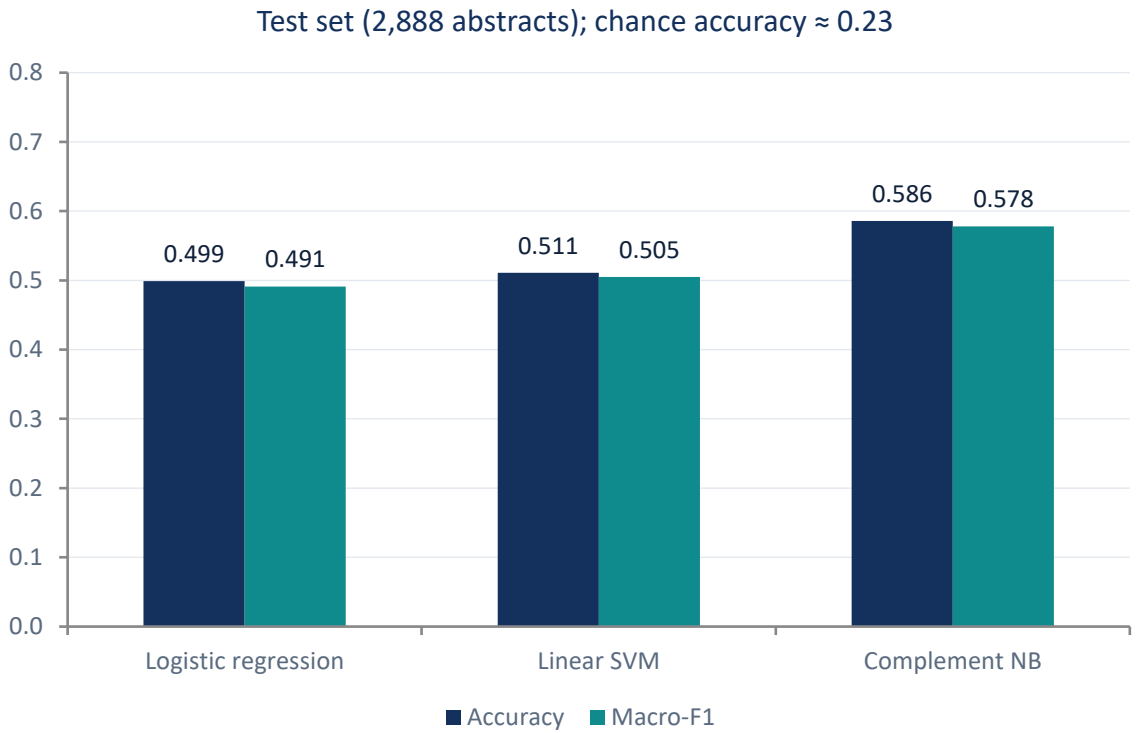
**INPUT**
Abstract text

**PROCESSING**
spaCy tokenise + lemmatise → TF-IDF → classifiers; SVD + k-means

**OUTPUT**
Predictions, metrics, clusters

**METHODS IMPLEMENTED**

- TF-IDF: unigrams + bigrams, sublinear TF, min_df 2
- Logistic regression, linear SVM, Complement Naive Bayes
- Truncated SVD (100-d) → k-means; silhouette, ARI, NMI, purity




**ACTUAL RESULT**

Best: Complement NB accuracy 0.586, macro-F1 0.578 (chance ≈ 0.23). k = 5 clusters: NMI 0.108

**KEY LEARNING**

Macro-F1 weights every class equally; overlapping classes cap accuracy

**IN PNEUMOCARE**

Clinical Text → Classification / Clustering

# T11 · Clinical Information Extraction and Entity Recognition

*Extract problems, medications with doses, vitals, labs and procedures from free text, including negation*

**DATASET** REAL + SYNTHETIC

3 synthetic notes written for the project; 200 Medical Abstracts for corpus statistics

**WORKFLOW**

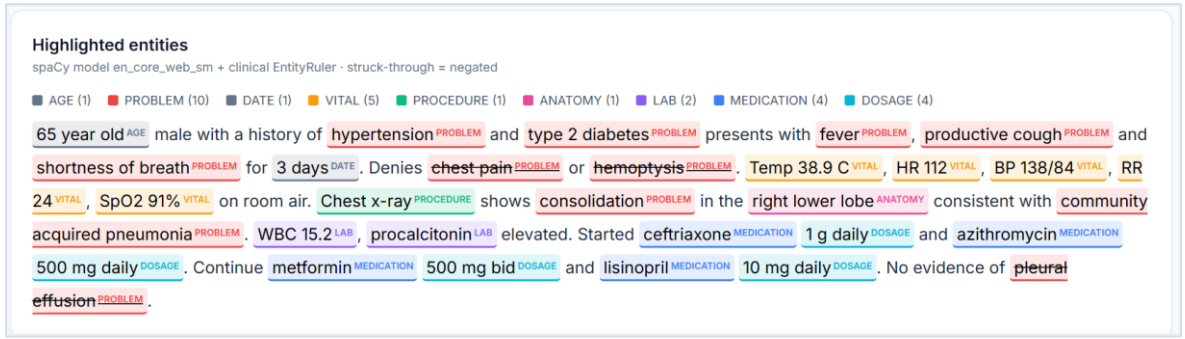
**INPUT**
Clinical note

**PROCESSING**
spaCy en_core_web_sm + clinical EntityRuler + negation rules

**OUTPUT**
Highlighted entities + structured JSON

**METHODS IMPLEMENTED**

- Statistical NER plus a rule lexicon (~80 problems, 44 medications)
- Token patterns for dosage, vitals, lab values, age
- NegEx-style negation (6-token window); history and family flags




| | |
|---|---|
| **Negated** | chest pain, hemoptysis, pleural effusion |
| **Medications** | ceftriaxone 1 g daily · azithromycin 500 mg daily · metformin 500 mg bid · lisinopril 10 mg daily |
| **Vitals** | Temp 38.9 C · HR 112 · BP 138/84 · RR 24 · SpO2 91% |

**ACTUAL RESULT**
ED note: 29 entities; 3 negations correct; all 4 medications linked to doses; 5 vitals extracted

**KEY LEARNING**
General NER misses clinical concepts; rules + negation context turn notes into structured data

**IN PNEUMOCARE**
Clinical Text → Entity extraction

# T12 · Clinical Text Ranking and Word Disambiguation

*Rank documents by relevance to a query, and resolve ambiguous clinical abbreviations from context*

**DATASET** REAL + SYNTHETIC

Ranking: 11,550 Medical Abstracts (real). WSD: 43 hand-written labelled sentences, 6 abbreviations

**WORKFLOW**

**INPUT**
Query, or a sentence with an abbreviation

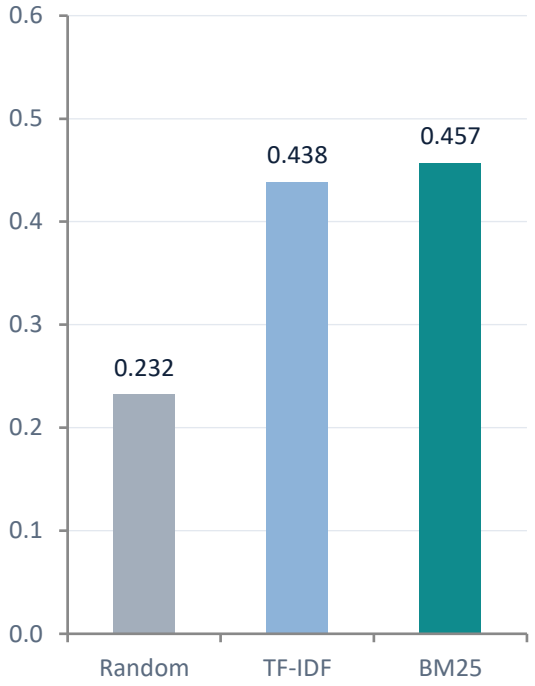
**PROCESSING**
TF-IDF cosine & BM25; simplified Lesk overlap

**OUTPUT**
Ranked list, P@10 / MAP; chosen sense

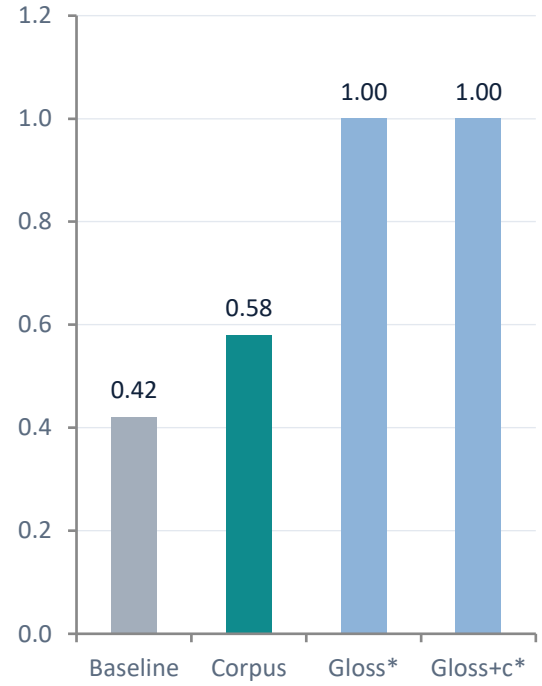
**METHODS IMPLEMENTED**

- Vector space model (cosine) and BM25 (k1 1.5, b 0.75)
- 200 test titles as queries; same label = relevant (proxy)
- Simplified Lesk: gloss, gloss + corpus, corpus-only signatures

Retrieval precision@10




WSD accuracy (43 sentences)




**ACTUAL RESULT**
BM25 P@10 0.457 vs random 0.232. Corpus-only Lesk 0.58 vs 0.42 (*gloss 1.00 is optimistic)

**KEY LEARNING**
Retrieval needs an evaluation protocol; disambiguation depends on context words

**IN PNEUMOCARE**
Clinical Text → Ranking & disambiguation

# T13 · Dimensionality Reduction for EDA

*Reduce many correlated variables to a few components and use them for exploratory analysis*

**DATASET** REAL DATA

NHANES 2017–2018 adults: 15 numeric variables, n = 5,120 (also the Synthea patient profile)

**WORKFLOW**

**INPUT**
Numeric features

→

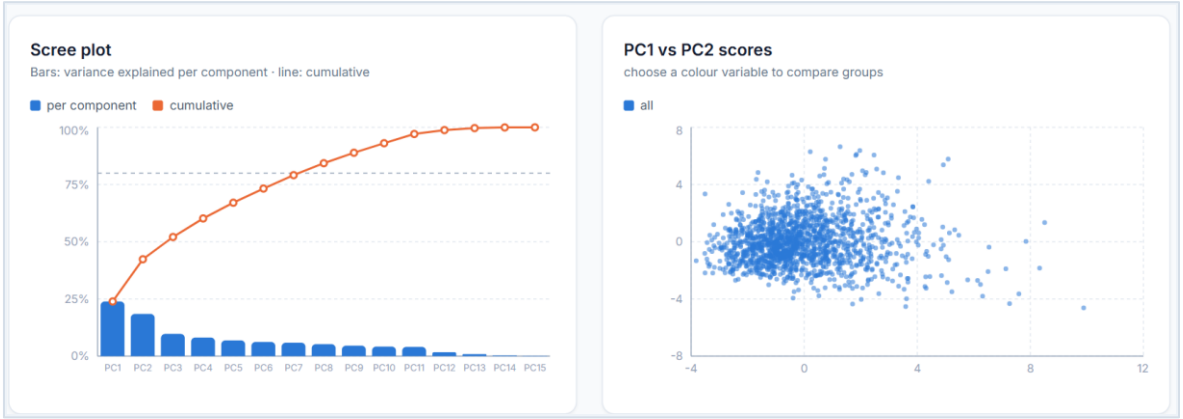
**PROCESSING**
Median impute → z-score → PCA

→

**OUTPUT**
Scree, loadings, 2-D/3-D scores, correlation map

**METHODS IMPLEMENTED**

- Scaling compared: z-score, min-max, centring only
- Explained variance and cumulative curve (80% line)
- Loadings interpret components; scores coloured by group




**PC1 · diet quantity**
energy +0.49, fat +0.45, protein +0.42

**PC2 · body size**
waist, BMI, weight ≈ +0.54

**PC3 · activity vs age**
activity +0.56, age −0.47, sedentary −0.34

**ACTUAL RESULT**
8 of 15 components explain 80%. PC1 holds 76% of variance unscaled vs 24% scaled

**KEY LEARNING**
Scale before PCA; components summarise correlation and are not causes

**IN PNEUMOCARE**
Advanced Analytics → PCA & EDA

# T14 · Behavioral Factors Influencing Adult Weight Part 1

*Describe adult BMI and lifestyle factors and how they relate*

**DATASET** REAL DATA

NHANES 2017–2018: 7 files merged on SEQN; adults ≥ 20, not pregnant, BMI measured (n = 5,120)

**42.5%**

obesity, survey-weighted (CDC ≈ 42%)

**41.8%**

obesity, unweighted sample

**WORKFLOW**

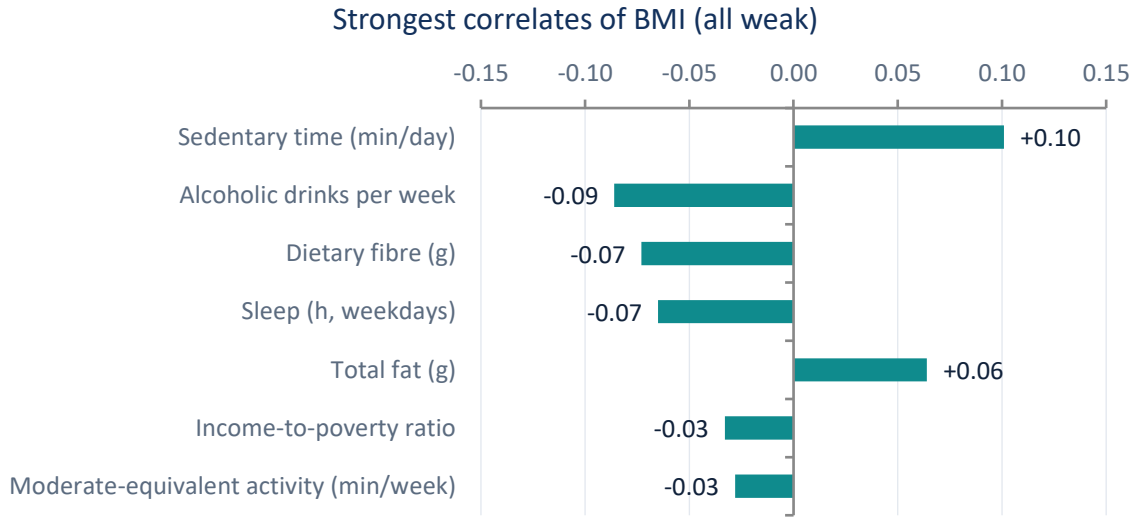
**INPUT**
7 NHANES survey files

**PROCESSING**
Merge, recode survey codes, derive activity, alcohol, smoking

**OUTPUT**
BMI categories, weighted prevalence, correlations, tests

**METHODS IMPLEMENTED**

- Codes 7/9/77/99/… recoded as missing; vigorous minutes ×2
- WHO BMI categories; survey weight WTMEC2YR
- Spearman correlation; Kruskal–Wallis across groups




**ACTUAL RESULT**

Weighted obesity 42.5% (CDC ≈ 42%). Top correlate: sedentary time ρ = +0.10; inactive BMI 30.3 vs 29.6

**KEY LEARNING**

Survey data need recoding and weights; correlations here are weak and not causal

**IN PNEUMOCARE**

Advanced Analytics → Adult weight: exploration

# T15 · Behavioral Factors Influencing Adult Weight Part 2

*Model BMI (regression) and obesity (classification) from behavioural and demographic factors*

**DATASET** REAL DATA

NHANES adults from T14 (n = 5,120): 11 numeric + 4 categorical predictors; 80/20 split

**WORKFLOW**

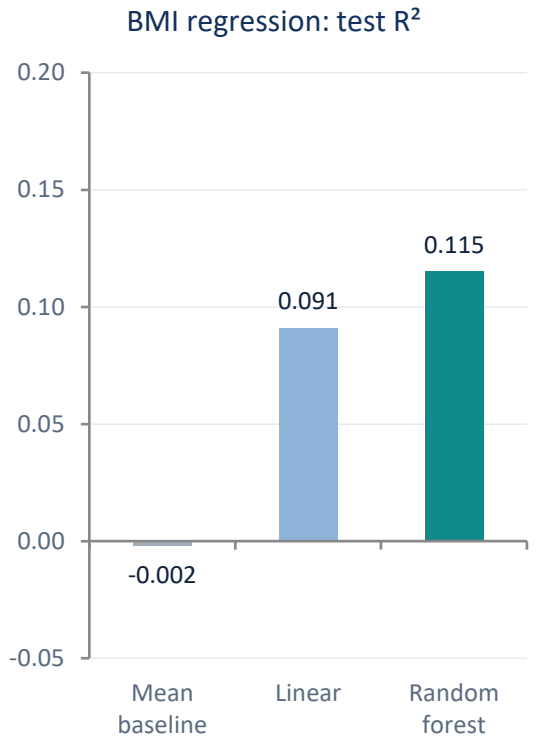
**INPUT**
Lifestyle and demographic predictors

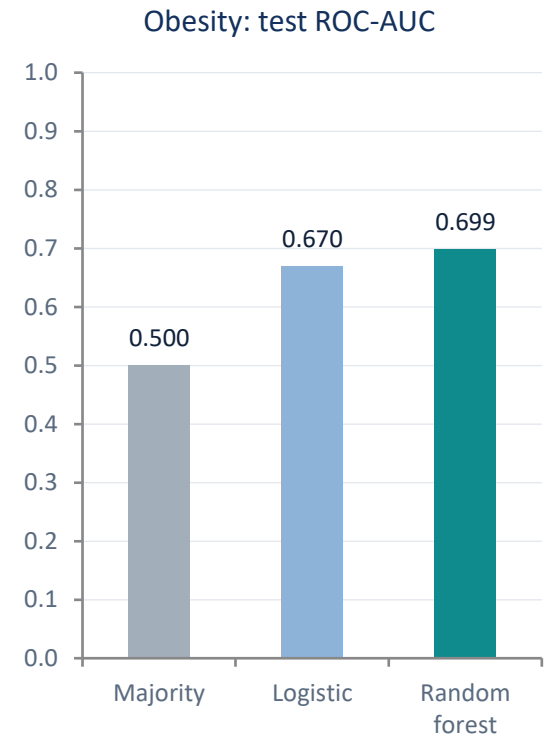
**PROCESSING**
Impute, scale, one-hot; 4 models; 5-fold CV

**OUTPUT**
Metrics, importance, coefficients, odds ratios

**METHODS IMPLEMENTED**

- Weight, height, waist excluded (BMI leakage)
- Linear regression & random forest (BMI); logistic & RF (obese)
- Permutation importance, coefficients, odds ratios




**ACTUAL RESULT**
RF $R^2$ 0.115 (CV 0.100); obesity AUC 0.699, accuracy 0.650 vs majority 0.582

**KEY LEARNING**
Lifestyle factors explain ~10% of BMI variation: associated, but weak individual predictors

**IN PNEUMOCARE**
Advanced Analytics → Adult weight: models

# Testing and verification

**85**
backend tests passed (pytest)

**15**
frontend tests passed (Vitest + RTL)

**15/15**
tutorials verified via live API (64/64 checks)

**9/9**
independent model-audit checks

**46/46**
live API contract checks: errors, uploads, 24 exports

**0**
lint (ruff) and type-check (tsc) errors; build OK

### WHAT THE VERIFICATION ACTUALLY CHECKS

- Each tutorial's workflow runs on its intended dataset through the API, and outputs are checked (e.g. best model = lowest MAE, variance sums to 1)
- Model metrics recomputed with plain NumPy/scikit-learn, independent of the project's metric code
- Every page and button was clicked through in a scripted browser walkthrough (28 Oct); there is no automated browser test suite, and no load or security testing

# Limitations and critical observations

### Grad-CAM does not localise to the lungs

On 156 test images, Grad-CAM mass inside the lung fields (0.304) ≈ their area share (0.316). The model may use non-lung cues.

### Threshold does not transfer

Validation threshold 0.029 gives specificity 0.816 on test (0.949 at 0.5). Needs calibration, never test-set tuning.

### Dataset scope

Paediatric images only, down-sampled; no external validation; no patient-level leakage check for MedMNIST.

### Small or synthetic data

Synthea operations data have 111 patients; WSD and NER notes are hand-written and synthetic.

### Classical segmentation

T4 lung-field segmentation has no ground truth and misses consolidated lung; checked only for plausibility.

### Text and survey models

Abstracts are not clinical notes; scispaCy not installed; T10 untuned; T14–T15 lack survey-design errors.

# Future improvements (prioritised)

| | | |
|---|---|---|
| 1 | **Calibrate probabilities** | Temperature scaling or threshold selection on deployment-like validation data; report calibration curves |
| 2 | **Test where the model looks** | Validated lung masks (a pretrained anatomical model checked against ground truth) and lung-cropped training |
| 3 | **External validation** | An independent dataset from a different population; the Kaggle copy is the same patients at higher resolution |
| 4 | **Browser end-to-end tests** | Playwright tests for upload, training and Image Lab workflows |
| 5 | **Stronger text models** | Install scispaCy clinical NER; tune T10 with cross-validation on training data only |
| 6 | **Survey-correct statistics** | Design-based standard errors and weighted models for T14–T15 |